

Adshaya Balarajah

Software Engineer

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🐙 Github | 📁 Portfolio

PROFILE

Ambitious software engineer with hands-on experience designing, building, and deploying scalable full-stack applications. Skilled in backend systems and intuitive frontends, with a strong emphasis on clean code, system reliability, and measurable impact. Eager to join a high-performing engineering team to contribute to technically challenging, high-impact products.

EDUCATION

- **University of Moratuwa, Colombo, Sri Lanka** (May 2022 – Present)
B.Sc. (Hons) in Information Technology SGPA L2S2: **3.0/4.0**
- **V/Vavuniya Tamil Madhya Maha Vidyalaya, Sri Lanka** (2010 – 2018)
High School | Bio Science: 2B's and C

TECHNICAL SKILLS

- **Languages:** Java, C, JavaScript, Python
- **Frontend:** React, React Native, HTML, CSS, Tailwind CSS, Material UI, SASS
- **Backend:** Spring Boot, FastAPI
- **Databases:** MySQL, PostgreSQL
- **Tools:** Git, VS Code, IntelliJ, Postman, Figma

WORK EXPERIENCE

- **Software Engineer Intern** Mar 2025 – Sep 2025
Techorin Solutions (Pvt) Ltd
GeniusBees — AI-enhanced educational platform with gamified learning and interactive quizzes.
 - **UI/UX Design:** Designed end-to-end interfaces in Figma for authentication pages, admin panels, and student/parent dashboards.
 - **Frontend Development:** Implemented responsive login and sign-up pages using React and Tailwind CSS from Figma prototypes.
 - **Template Development:** Built interactive drag-and-drop, image sorting, and multiple-selection educational templates with audio integration.
 - **Client Collaboration:** Participated in daily stand-ups and weekly demo sessions, refining designs based on client feedback.

RESEARCH

- **Multimodal Emotional Recognition System with Explainability (MedOracle)**
Final Year Project
Technologies: React.js, Python, PyTorch, PostgreSQL, OpenCV, FastAPI
 - **SHAP Explainability:** Applied Kernel SHAP to the fusion pipeline to compute per-modality feature importance with a quantitative faithfulness metric.

- **LLM Chatbot:** Built an Anthropic Claude API-powered chatbot converting SHAP outputs into plain-English clinical explanations with multi-turn Q&A.
- **FastAPI Backend:** Implemented 9 REST API endpoints with JWT authentication, bcrypt hashing, and full session management.
- **Database Design:** Architected a PostgreSQL database schema across 4 tables using SQLAlchemy ORM and Alembic migrations.
- **React Dashboard:** Built an interactive dashboard with emotion trend charts, SHAP bar charts, and integrated chatbot panel using Recharts and Axios.

PROJECTS

- **Labour Hiring System (LabourLINK)** - 2nd Year Software Project

[Github](#) 

Technologies: React.js, React Native, Spring Boot, MySQL | **Client:** AlphaCodes (Pvt) Ltd

- Developed web and mobile applications for on-demand labor hiring with map integration and real-time communication.
- Built mobile labor dashboard for profile updates, booking management, review viewing, reporting, and account management.
- Contributed to customer dashboard displaying user profiles, job categories, top-rated employees, and booking details.
- Ensured successful API integration with external systems and real-time data updates using Postman for API testing.
- Improved user interface elements across the project using React Native Paper.

- **Food Delivery App (Foodie App)** - Individual Project

[Github](#) 

Technologies: React.js, Spring Boot, MySQL, Tailwind CSS, MUI

- Developed a full-stack multi-vendor online food delivery platform enabling restaurant owners to register and customers to browse, order, and pay seamlessly.
- Built dedicated restaurant admin panels and a super admin panel for platform-wide management.
- Implemented customer-facing features including Add to Favourite, Carousel Display, Add to Cart, and order tracking.
- Integrated JWT, Spring Security, Stripe payment gateway, and email notifications for secure operations.
- Designed and developed RESTful APIs using Spring Boot 3 to support multi-vendor operations and real-time data flow.

- **Automated Customized Popcorn Maker** - 1st Year Hardware Project

Technologies: C, Arduino IDE, Arduino Mega 2560

- Developed an efficient popcorn maker using Arduino Mega to automate the process, reducing manual intervention and minimizing manpower.
- Integrated DS18B20 temperature sensor for heating element control and MQ2 gas sensor for smoke detection with automatic exhaust fan control.

ACHIEVEMENT & PARTICIPATION

- HackMoral 6.0 — INTECS, Faculty of Information Technology, University of Moratuwa